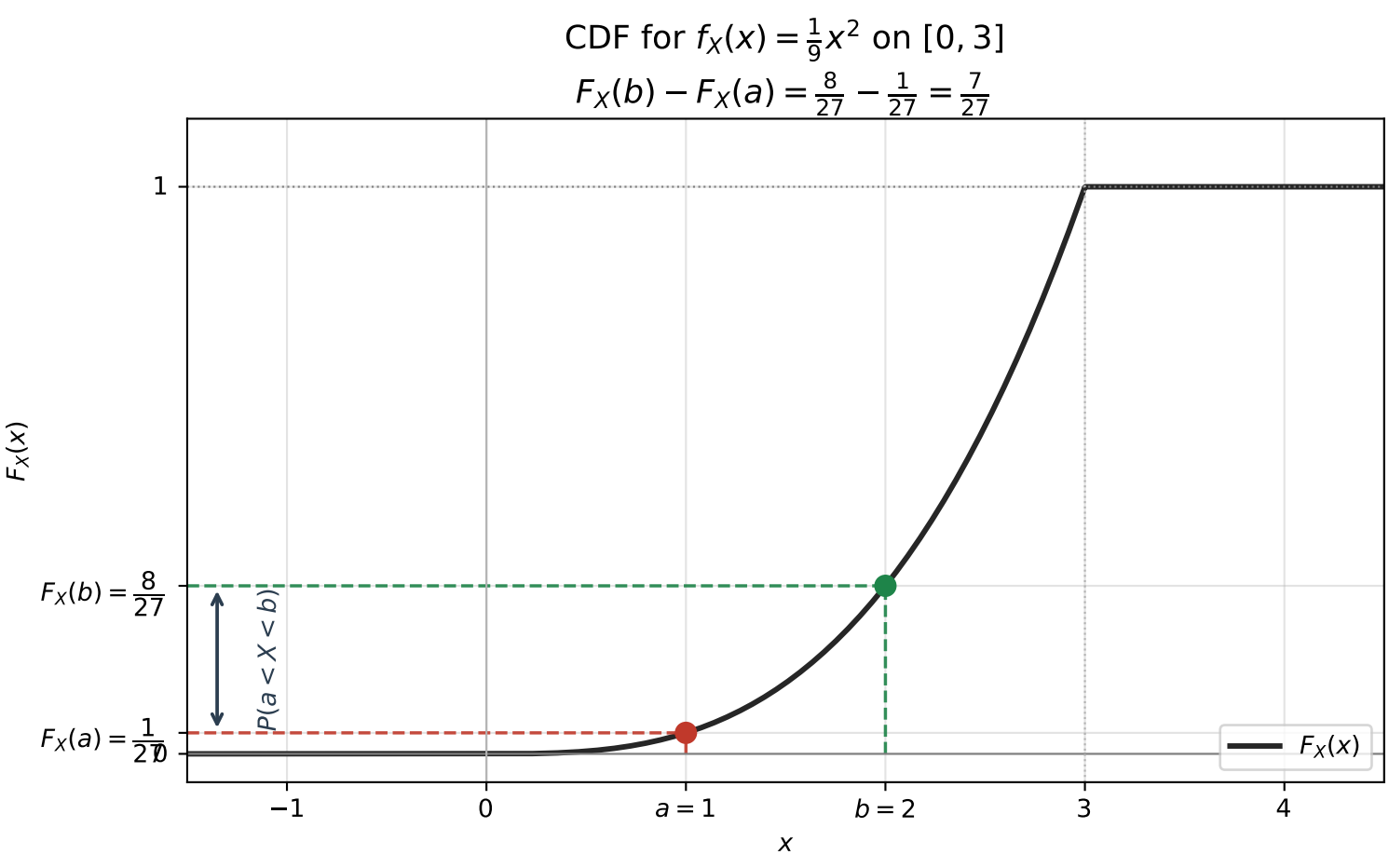
CDF for $f_X(x) = \frac{1}{9}x^2$ on $[0, 3]$
$F_X(b) - F_X(a) = \frac{8}{27} - \frac{1}{27} = \frac{7}{27}$
1
$F_X(x)$
$F_X(b) = \frac{8}{27}$
$P(a < X < b)$
$F_X(a) = \frac{1}{27}$
$F_X(x)$
−1
0
$a = 1$
$b = 2$
3
4
x